

SMARTBURSARY

Intelligent Bursary Application, Verification & Tracking System

Software Requirements Specification (SRS)

Version 1.0

Project Domain: Education / Financial Assistance

System Type: Web / Mobile Information System

Core Technologies: Web/Mobile, Database, AI-assisted Verification

Academic Group Project

1. Introduction

1.1 Project Overview

SmartBursary is a digital bursary management system designed to simplify and improve the process of applying for, reviewing, verifying, approving and monitoring bursary applications. Applicants receive a convenient platform for submitting applications and tracking progress, while administrators and committee members receive a centralized platform for verification, evaluation, reporting and decision support. The system incorporates AI-assisted application analysis and duplicate/fraud-risk detection to help authorized reviewers process large numbers of applications efficiently.

2. Problem Statement

Traditional bursary application processes may depend heavily on paper forms, physical submissions, manual verification and fragmented records. These approaches can lead to long processing times, incomplete or lost applications, duplicate submissions, difficult verification, limited status visibility, heavy administrative workloads and poor record management. SmartBursary addresses these challenges through a centralized digital workflow that improves accessibility, efficiency, transparency, security and accountability.

3. Aim of the Project

To develop a secure and user-friendly digital bursary management system that enables applicants to submit and track applications while allowing administrators to efficiently verify, evaluate and manage bursary applications.

4. Objectives

- Digitize the bursary application process and reduce paperwork.
- Allow applicants to submit applications remotely and upload supporting documents.
- Provide application reference numbers and status tracking.
- Enable administrators to review applications through a centralized dashboard.
- Detect possible duplicate applications and suspicious records.
- Provide AI-assisted application prioritization and verification support.
- Improve protection of applicant and financial information.
- Generate reports and statistics for authorized administrators.
- Improve communication between applicants and bursary administrators.
- Maintain reliable historical records of applications and decisions.

5. Scope of the System

5.1 In Scope

- Applicant registration, authentication and profile management.
- Creation, editing and submission of bursary applications.
- Viewing available bursary programs and eligibility criteria.
- Secure upload and management of supporting documents.
- Application validation, review, verification and status tracking.
- AI-assisted prioritization, document analysis and duplicate-risk detection.
- Administrator, reviewer and system-administrator dashboards.
- Notifications, audit logs, search, filtering and reporting.

6. Users and Roles

Applicant: Registers, completes a profile, applies for bursaries, uploads documents, submits applications, tracks status and receives notifications.

Administrator: Manages users and bursary programs, reviews applications, assigns reviewers, verifies information, manages decisions and generates reports.

Reviewer / Committee Member: Reviews assigned applications and documents, sees AI-assisted indicators, adds comments and submits recommendations.

System Administrator: Manages roles, permissions, security settings, audit logs, backups and technical configuration.

7. Functional Requirements

FR-001 User Registration: The system shall allow applicants to create accounts using required personal information and shall validate required fields and prevent duplicate accounts where applicable.

FR-002 Authentication: The system shall provide secure login/logout and enforce role-based access to protected functions.

FR-003 Applicant Profile: Applicants shall maintain personal, educational, family and financial information required by the selected bursary.

FR-004 Available Bursaries: Applicants shall view active bursary programs, descriptions, eligibility criteria, deadlines, funding information and required documents.

FR-005 Create Application: Applicants shall create and save applications as drafts before submission.

FR-006 Application Validation: Before submission, the system shall check required information and documents and identify missing items.

FR-007 Application Submission: On successful submission, the system shall generate a unique application number, record the submission date, set status to Submitted and notify the applicant.

8. Document Management

- Applicants may upload identification, student, academic, financial and other required documents.
- The system shall validate supported formats and file-size limits.
- Documents shall be associated with the correct application and stored securely.
- Authorized reviewers shall be able to view and verify documents.
- The system shall identify missing required documents.

9. AI-Assisted Features

9.1 AI Application Analysis

The system may analyze application information and identify indicators such as financial need, household size, educational need, completeness and possible duplicate information. It can generate an assistance-priority score to help reviewers decide which applications may require earlier attention.

Example: Financial Need: HIGH; Educational Need: HIGH; Application Complete: YES; Duplicate Risk: LOW; Priority Score: 87%; Recommendation: HIGH PRIORITY FOR HUMAN REVIEW.

9.2 Duplicate Detection

The system shall identify potentially duplicated applications using configurable matching indicators such as identification number, student number, phone number, applicant information, guardian information and institution. Suspected matches shall be flagged for human review rather than automatically rejected.

9.3 Document Analysis

Where technically feasible, OCR or document-analysis tools may extract information from uploaded documents and compare it with application data. Human verification remains necessary for final confirmation.

10. Human Decision Requirement

The AI shall not independently make final bursary-award decisions. AI outputs are decision-support indicators. Final approval or rejection remains under the control of the authorized bursary committee or administrator.

11. Application Review and Status

Authorized reviewers shall view applicant information, supporting documents, eligibility information, AI-assisted indicators, comments and recommendations. Applicants shall be able to track statuses including Draft, Submitted, Under Review, Verification, Additional Information Required, Committee Review, Approved, Rejected and Disbursed.

12. Notifications

- Application submitted.
- Application moved to review or verification.
- Additional information or documents required.
- Application approved or rejected.
- Funding/disbursement update where applicable.
- Notifications may be implemented through in-app messaging, email and optionally SMS.

13. Administrator Dashboard

The dashboard shall display application statistics such as total applications, pending applications, applications under review, approved applications and rejected applications. Administrators shall be able to search and filter applications by status, date, institution, bursary program, priority and other configured criteria.

14. Reporting Requirements

- Application totals and trends.
- Approved, rejected and pending applications.
- Applications by institution or bursary program.
- Funding allocation summaries.
- Verification and duplicate-risk statistics.
- Export to PDF and/or CSV/Excel where supported.

15. Search Requirements

Authorized users shall be able to search by application number, applicant name, student number, identification number, institution and status.

16. Security Requirements

- Secure password handling using strong password hashing.
- Role-based access control.
- Protection of sensitive data during transmission and storage.
- Restricted access to uploaded documents.
- Audit logging of important administrative and review actions.
- Secure session/token management and logout.

- Regular backups and appropriate recovery procedures.

17. Non-Functional Requirements

Performance: Normal operations should respond within an acceptable time under expected system load.

Availability: The service should be available whenever applications need to be submitted or monitored, excluding planned maintenance.

Usability: Interfaces should be simple, clear, mobile-friendly and understandable to users with limited technical experience.

Scalability: The architecture should support additional applicants, bursary programs, institutions and reviewers without major redesign.

Reliability: Submitted applications should be preserved and protected from accidental loss.

Maintainability: Components should be modular and documented so individual services can be updated independently.

Security: The system shall protect applicant information against unauthorized access, alteration and disclosure.

18. Proposed System Architecture

Applicant Web/Mobile Interface → Frontend → Backend API → Database / AI Services / Secure Document Storage

The frontend provides the user interface. The backend handles authentication, application processing, business rules and APIs. The database stores structured records. AI services provide decision-support analysis. Secure document storage holds uploaded supporting files.

19. Core Database Entities

Entity	Key information
Users	user_id, name, email, phone, password_hash, role, created_at
Applicants	applicant_id, user_id, identification, student number, institution, course, year, financial information
Bursaries	bursary_id, name, description, eligibility, deadline, funding amount, status
Applications	application_id, applicant_id, bursary_id, submission date, status, priority score, comments
Documents	document_id, application_id, type, file path, verification status
Reviews	review_id, application_id, reviewer_id, score, recommendation, comments
Notifications	notification_id, user_id, message, status, created_at
Audit Logs	log_id, user_id, action, timestamp, affected record

20. Eligibility Engine

Administrators shall be able to configure eligibility criteria for bursary programs. The system shall check whether submitted applications satisfy required conditions and identify missing requirements. Eligibility results should support, not replace, human review where the bursary rules require discretion.

21. Proposed Technology Stack

- Frontend: React, Next.js, or another suitable web/mobile framework.
- Backend: FastAPI/Python, Node.js/Express, Django, or another suitable API framework.
- Database: PostgreSQL or MySQL.
- AI: Python with suitable machine-learning, NLP and/or OCR libraries.
- Deployment: Appropriate cloud or local deployment environment.

22. System Workflow

Register/Login → Complete Profile → Select Bursary → Fill Application → Upload Documents → Validate → Submit → Duplicate Analysis → AI Assistance → Human Review → Additional Information / Approved / Rejected → Notification → Disbursement where applicable.

23. Expected Benefits

- Reduced paperwork and administrative workload.
- Faster and more organized application processing.
- Improved applicant convenience and status visibility.
- Improved document and record management.
- Reduced duplicate applications and improved data integrity.
- Better reporting and transparency.
- Decision-support tools for authorized reviewers.
- Improved communication with applicants.

24. Limitations

- The prototype may have limited access to real bursary datasets.
- AI accuracy depends on the quality and quantity of available data.
- External SMS and payment integrations may be unavailable during initial development.
- Some document verification will still require human review.
- Internet connectivity may affect access for some applicants.

25. Future Enhancements

- Dedicated Android/iOS mobile application.
- Integration with approved digital disbursement/payment services.
- Automated SMS notifications.
- Improved AI document and fraud-risk analysis.
- Identity-verification integrations where legally permitted.
- Advanced analytics and funding-demand prediction.
- Multi-organization support for different bursary providers.

26. Success Criteria

- Applicant registration and secure login work correctly.
- Applicants can create, save, submit and track applications.
- Required documents can be uploaded and reviewed.
- Administrators can manage bursary programs and applications.
- Reviewers can evaluate assigned applications.
- Duplicate-risk and AI-assisted indicators can be demonstrated.
- Role-based access prevents unauthorized functions.
- Notifications and reports can be demonstrated.
- Application records are securely stored and retrievable.

27. Conclusion

SmartBursary provides a centralized, secure and intelligent approach to bursary management. By digitizing applications and adding AI-assisted verification, duplicate detection and prioritization, the system can reduce administrative workload while improving accessibility, transparency and accountability. Human decision-makers remain responsible for final funding decisions. The project combines Software Engineering, database management, artificial intelligence, security and information systems to address a practical real-world problem.